

Supplementary material for *Anisotropic phonon-mediated electronic transport in chiral Weyl semimetals*

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(Dated: September 9, 2021)

SCATTERING LIFETIMES

We compute the lifetimes associated with four types of microscopic scattering events in NbGe₂ and NbSi₂: electron-phonon scattering ($\tau_{\text{e-ph}}$), momentum-relaxing electron-phonon scattering ($\tau_{\text{e-ph}}^{\text{MR}}$), electron-electron scattering mediated by the Coulomb interaction ($\tau_{\text{e-e}}$), and phonon-mediated electron-electron scattering ($\tau_{\text{e-e}}^{\text{ph}}$). Our methodology, for the most part, follows that presented in previous works [1–4]

We calculate the temperature-dependent, state-resolved electron-phonon scattering rate using Fermi’s golden rule, as presented in Refs. 1–4:

$$\frac{1}{\tau_{\text{e-ph}}(\mathbf{k}, n)} = \frac{2\pi}{\hbar} \int_{\text{BZ}} \frac{\Omega d\mathbf{k}'}{(2\pi)^3} \sum_{n'\alpha\pm} \delta(\varepsilon_{n'\mathbf{k}'} - \varepsilon_{n\mathbf{k}} \mp \hbar\omega_{\alpha\mathbf{k}'-\mathbf{k}}) \times \left[n_{\alpha\mathbf{k}'-\mathbf{k}} + \frac{1}{2} \mp \left(\frac{1}{2} - f_{n'\mathbf{k}'} \right) \right] \left| g_{n'\mathbf{k}',n\mathbf{k}}^{\alpha\mathbf{k}'-\mathbf{k}} \right|^2 \quad (\text{S1})$$

where Ω is the unit cell volume, $\varepsilon_{n\mathbf{k}}$ and $f_{n\mathbf{k}} = f(\varepsilon_{n\mathbf{k}}, T)$ are energies and Fermi occupations for an electron in band n at wave vector $\mathbf{k}$, $\hbar\omega_{\alpha\mathbf{q}}$ and $n_{\alpha\mathbf{q}} = n(\hbar\omega_{\alpha\mathbf{q}}, T)$ are energies and Bose occupations for a phonon with polarization index α at wave vector $\mathbf{q}$, and $g_{n'\mathbf{k}',n\mathbf{k}}^{\alpha\mathbf{q}}$ is the electron-phonon coupling matrix element:

$$g_{n'\mathbf{k}',n\mathbf{k}}^{\alpha\mathbf{q}} = \left(\frac{\hbar}{2m_0\omega_{\alpha\mathbf{q}}} \right)^{1/2} \langle \psi_{n'\mathbf{k}'} | \partial_{\alpha\mathbf{q}} V | \psi_{n\mathbf{k}} \rangle. \quad (\text{S2})$$

By momentum conservation, only processes where $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ are allowed, and this is already substituted into (S1). Phonon emission and absorption processes are accounted for in (S1) with the sum over $\pm$. The momentum relaxing (MR) scattering rate is identical to (S1) except for an additional factor accounting for the change in momentum between final and initial states based on their relative scattering angle:

$$\frac{1}{\tau_{\text{e-ph}}^{\text{MR}}(\mathbf{k}, n)} = \frac{2\pi}{\hbar} \int_{\text{BZ}} \frac{\Omega d\mathbf{k}'}{(2\pi)^3} \sum_{n'\alpha\pm} \delta(\varepsilon_{n'\mathbf{k}'} - \varepsilon_{n\mathbf{k}} \mp \hbar\omega_{\alpha\mathbf{k}'-\mathbf{k}}) \times \left[n_{\alpha\mathbf{k}'-\mathbf{k}} + \frac{1}{2} \mp \left(\frac{1}{2} - f_{n'\mathbf{k}'} \right) \right] \left| g_{n'\mathbf{k}',n\mathbf{k}}^{\alpha\mathbf{k}'-\mathbf{k}} \right|^2 \left(1 - \frac{v_{n\mathbf{k}} \cdot v_{n'\mathbf{k}'}}{|v_{n\mathbf{k}}| |v_{n'\mathbf{k}'}|} \right), \quad (\text{S3})$$

where $v_{n\mathbf{k}}$ is the group velocity. The resulting lifetimes for all six transition metal ditetrelides discussed are presented in Fig. S1.

To put the momentum-relaxing scattering into perspective, we also calculate lifetimes for electron-electron scattering mediated by both the Coulomb interaction and virtual phonons. The former is obtained by evaluating

$$\frac{1}{\tau_{\text{e-e}}(\mathbf{k}, n)} = 2 \int_{\text{BZ}} \frac{d\mathbf{k}'}{(2\pi)^3} \sum_{n'} \sum_{\mathbf{G}\mathbf{G}'} \tilde{\rho}_{n'\mathbf{k}',n\mathbf{k}}(\mathbf{G}) \tilde{\rho}_{n'\mathbf{k}',n\mathbf{k}}^*(\mathbf{G}') \times \frac{4\pi e^2}{|\mathbf{k}' - \mathbf{k} + \mathbf{G}|^2} \text{Im}[\epsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{k}' - \mathbf{k}, \varepsilon_{n\mathbf{k}} - \varepsilon_{n'\mathbf{k}'})], \quad (\text{S4})$$

where $\tilde{\rho}_{n'\mathbf{k}',n\mathbf{k}}(\mathbf{G})$ is the plane wave expansion of the product density $\sum_{\sigma} u_{n'\mathbf{k}'(\mathbf{r})}^{\sigma*} u_{n\mathbf{k}}^{\sigma}(\mathbf{r})$ of the Bloch functions with reciprocal lattice vectors $\mathbf{G}$, and $\epsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{k}' - \mathbf{k}, \varepsilon_{n\mathbf{k}} - \varepsilon_{n'\mathbf{k}'})$ is the microscopic dielectric function in a plane wave basis calculated within the random-phase approximation. The electron-electron lifetime (S4) is analytically extended to finite temperature values using a Fermi-liquid theory fit to the state-resolved result at zero temperature, i.e. by fitting, [1]

$$\tau_{\text{e-e}}^{-1}(\varepsilon, T) \approx \frac{D_e}{\hbar} [(\varepsilon - \varepsilon_F)^2 + (\pi k_B T)^2], \quad (\text{S5})$$

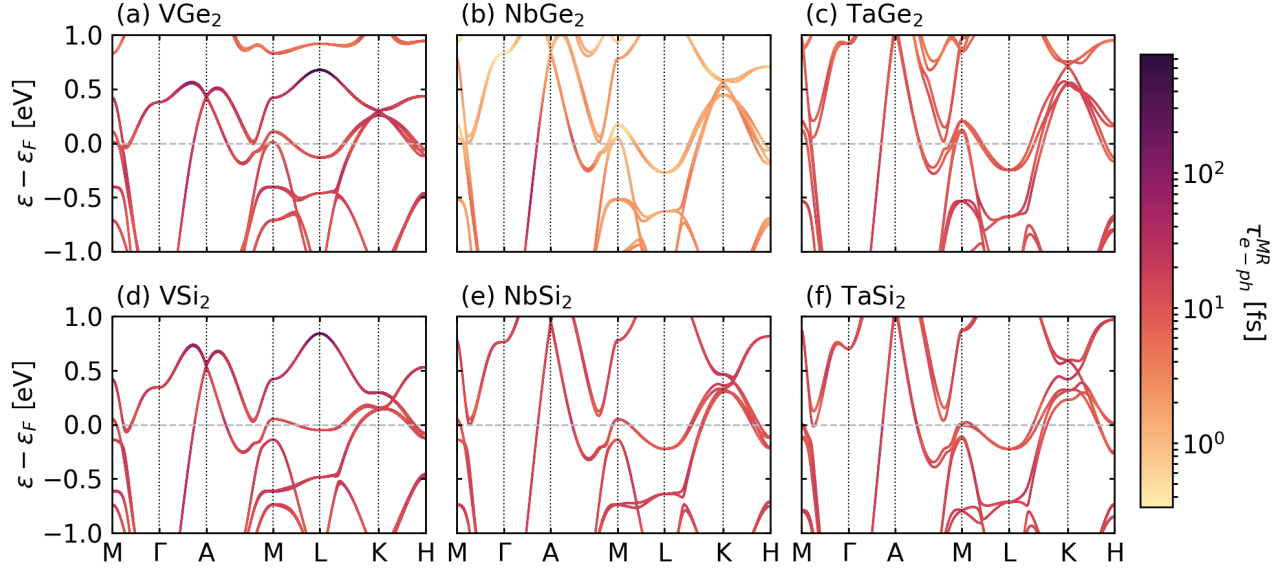


Fig. S1. Computed room temperature momentum-relaxing electron-phonon lifetimes from Eq. S3 plotted on the electronic band structures for all six transition metal ditellurides that crystallize in space group 180 or 181.

where ε_F is the Fermi energy.

We evaluate the phonon-mediated electron-electron scattering time as a Fermi-surface averaged version of Fermi's golden rule with a second-order electron-phonon matrix element according to

$$\frac{1}{\tau_{e-e}^{\text{ph}}} = \frac{2\pi}{\hbar g(\varepsilon_F)} \sum_{\alpha} \int \frac{\Omega d\mathbf{q}}{(2\pi)^3} G_{\mathbf{q}\alpha}^2 \gamma(\beta\hbar\bar{\omega}_{\alpha\mathbf{q}}), \quad (\text{S6})$$

where $\beta^{-1} = k_B T$, the Boltzmann constant k_B times the temperature T , $\bar{\omega}_{\alpha\mathbf{q}} = \omega_{\alpha\mathbf{q}}(1 + i\pi G_{\mathbf{q}\alpha})$ and we've introduced the complex-valued function $\gamma(x)$:

$$\gamma(x) \equiv \int_{-\infty}^{\infty} \frac{dy}{\sinh^2\left(\frac{y}{2}\right) |x - y|^2}. \quad (\text{S7})$$

The Fermi-surface averaged electron-phonon interaction strength can be obtained from

$$G_{\mathbf{q}\alpha} = \sum_{ab} \int \frac{g_s \Omega d\mathbf{k}}{(2\pi)^3} \left| g_{a\mathbf{k}, b\mathbf{k}+\mathbf{q}}^{\alpha\mathbf{q}} \right|^2 \delta(\varepsilon_{a\mathbf{k}} - \varepsilon_F) \delta(\varepsilon_{b\mathbf{k}+\mathbf{q}} - \varepsilon_F), \quad (\text{S8})$$

and constitutes the weight of each phonon mode in the Eliashberg spectral function:

$$\alpha^2 F(\omega) = \frac{1}{g(\varepsilon_F)} \sum_{\alpha} \int \frac{\Omega d\mathbf{q}}{(2\pi)^3} \delta(\omega - \omega_{\mathbf{q}\alpha}) G_{\mathbf{q}\alpha}. \quad (\text{S9})$$

We show $G_{\mathbf{q}\alpha}$ for both NbGe₂ (Fig. S2) and NbSi₂ (Fig. S3). Besides the orders of magnitude difference in the coupling between the two materials, it is remarkable that NbGe₂ shows a strong correlation between the direction of the phonon momenta and the coupling strength. In particular, modes that can contribute to electron scattering (in particular modes $\alpha = 12, \dots, 15$) with a change in momentum along the in-plane axis (out-of-plane axis) show particularly strong (weak) coupling, providing a partial explanation of the pronounced anisotropy in the resistivity (cf. Fig. 2(a) in the main text) compared to NbSi₂.

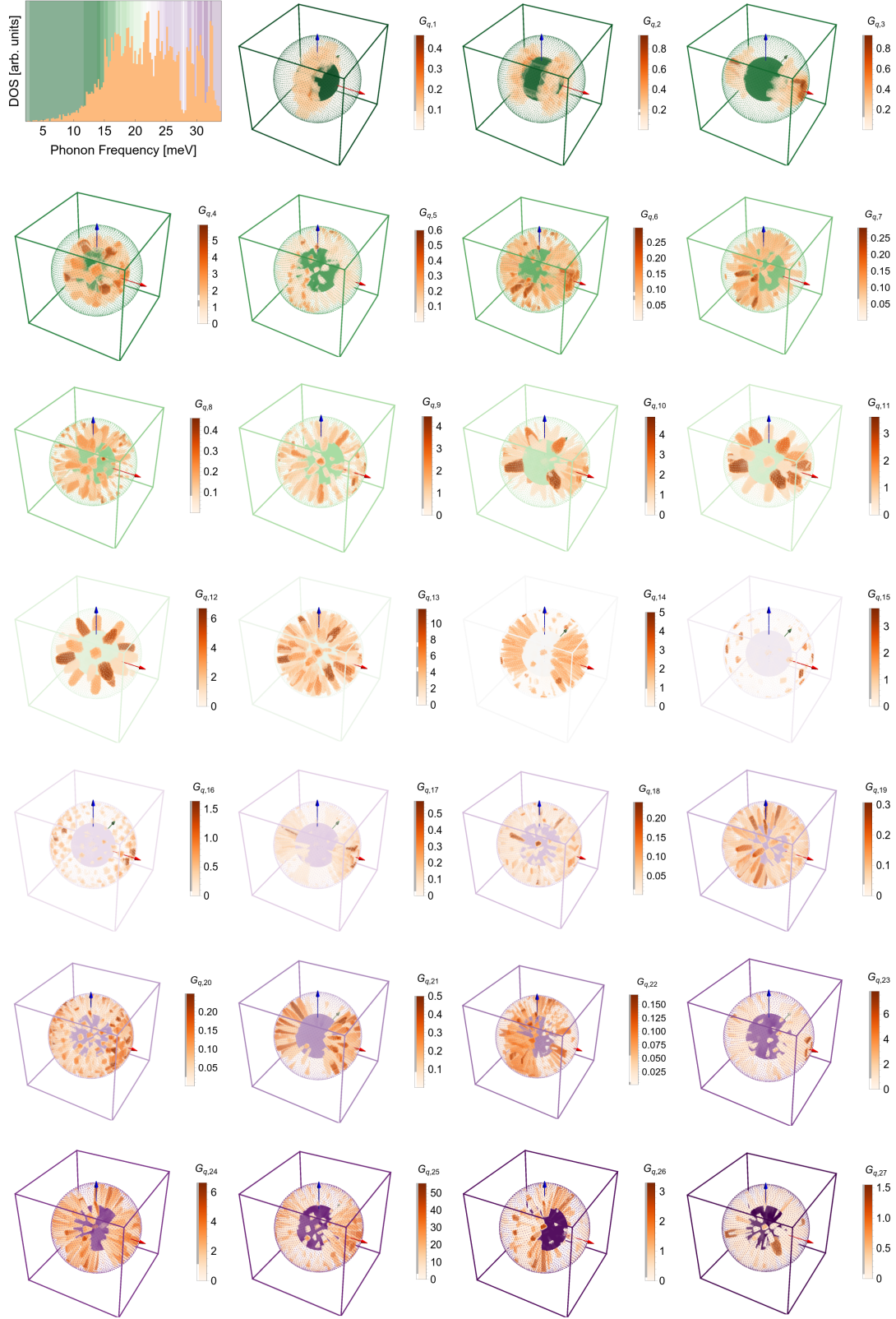


Fig. S2. Phonon density of states (orange) of NbGe₂ (top left panel), together with mode-resolved Fermi-surface averaged electron-phonon coupling strength (the energy window for each mode is given by the color-coded background rectangles in the top left panel from low (green) to high (purple), also reflected in the box frame). Subsequent panels show the angular-dependence in the phonon momentum q of the Fermi-surface averaged electron-phonon coupling strength G for various phonon modes α , inside a spherical shell given by the mode's energy window. The color-coding of the vectors represents the magnitude of G . Modes 10-12 exhibit strong in-plane anisotropy with strong electron-phonon couplings.

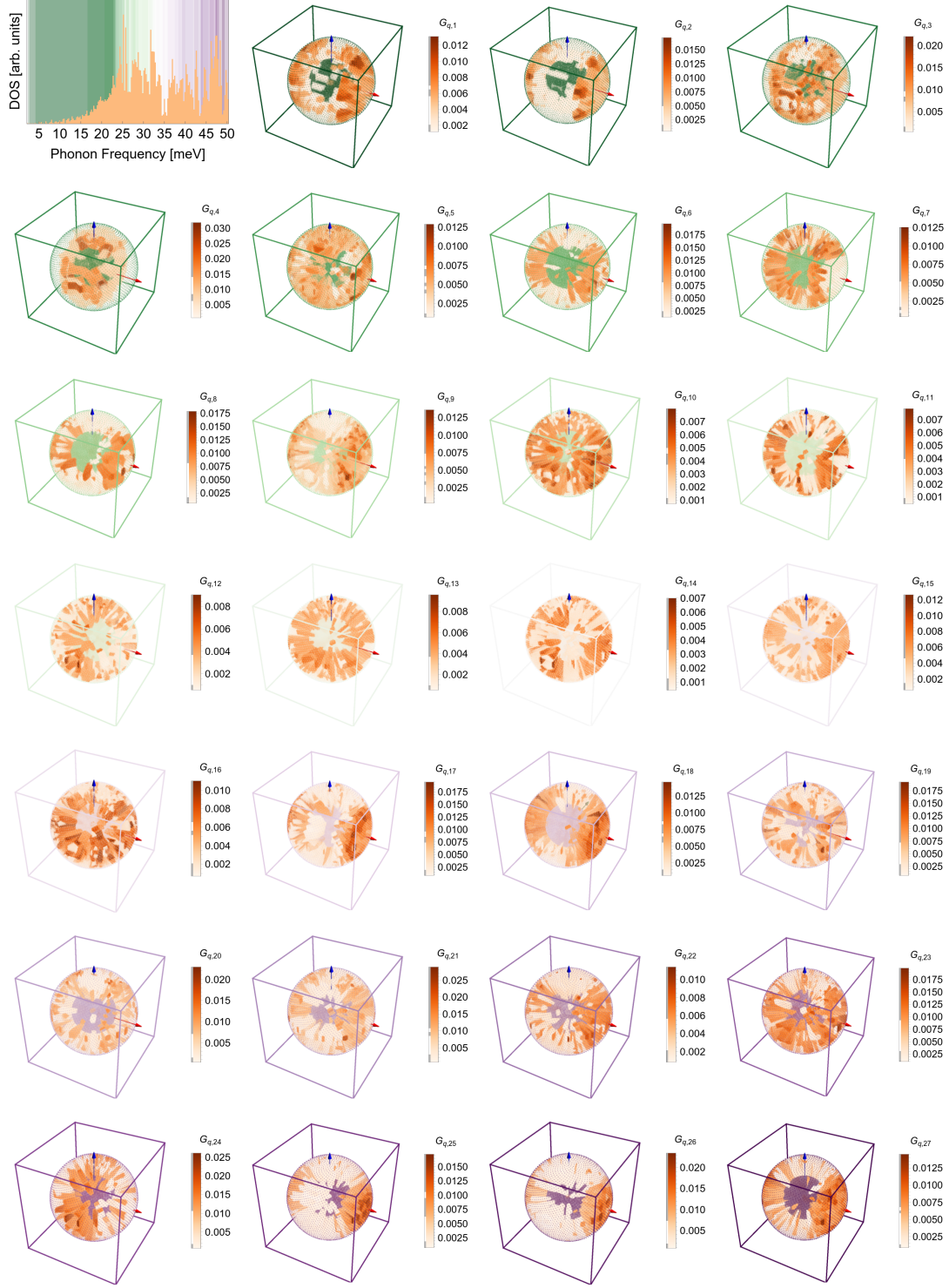


Fig. S3. Phonon density of states (orange) of NbSi₂ (top left panel), together with mode-resolved Fermi-surface averaged electron-phonon coupling strength (the energy window for each mode is given by the color-coded background rectangles in the top left panel from low (green) to high (purple), also reflected in the box frame). Subsequent panels show the angular-dependence in the phonon momentum q of the Fermi-surface averaged electron-phonon coupling strength G for various phonon modes α , inside a spherical shell given by the mode's energy window. The color-coding of the vectors represents the magnitude of G . In contrast to NbGe₂ (Fig. S2), electron-phonon coupling appears to be largely isotropic in NbSi₂

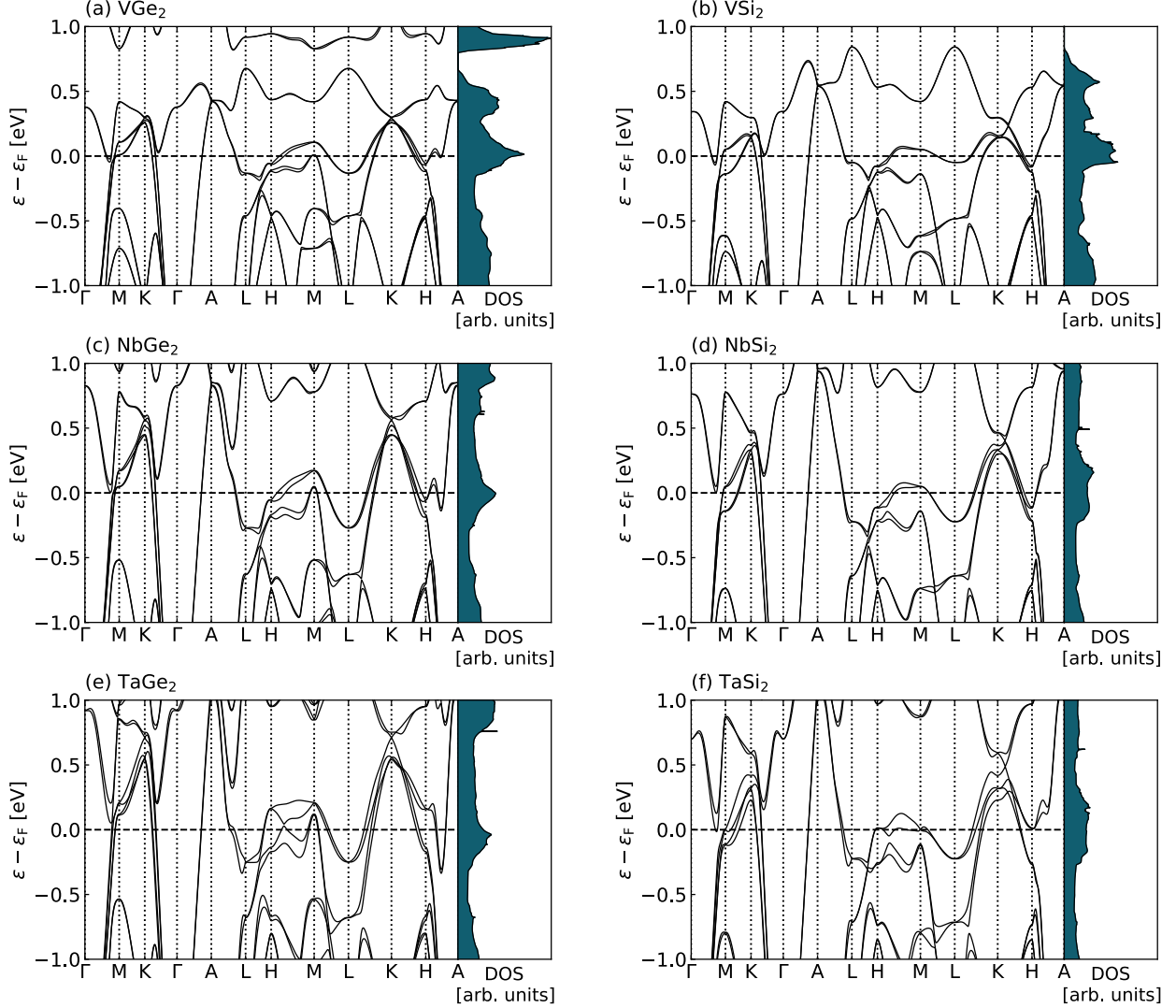


Fig. S4. *Ab initio* electron band structure and electron density of states close to the Fermi level for all six transition metal ditetrelides that crystallize in space group 180 or 181.

HYDRODYNAMIC PHASE DIAGRAM

We solve the electronic Boltzmann transport equation numerically, in the presence of momentum-conserving scattering. At steady-state, the evolution of the distribution function $f(\mathbf{r}, \mathbf{k})$ for non-equilibrium electrons in the neighborhood of position $\mathbf{r}$ with wave vector $\mathbf{k}$ is given by:

$$\mathbf{v}_{\mathbf{k}} \cdot \nabla_{\mathbf{r}} f(\mathbf{r}, \mathbf{k}) + e\mathbf{E} \cdot \nabla_{\mathbf{k}} f(\mathbf{r}, \mathbf{k}) = \Gamma[f], \quad (\text{S10})$$

where $\mathbf{v}_{\mathbf{k}}$ is the electron group velocity, and $\Gamma[f]$ is the collision integral. We consider flow through a 2D channel ($\mathbf{r} = (x, y)$) of width W , for a spherical Fermi surface ($\mathbf{v} = v_F(\cos \theta, \sin \theta)$), at the relaxation time approximation level. Under these approximations, we can linearize S10, to give an integro-differential equation in terms of an ‘effective’ mean free path, l_{eff} [5, 6]:

$$\sin(\theta) \partial_y l_{\text{eff}}(y, \theta) + \frac{l_{\text{eff}}(y, \theta)}{l} = 1 + \frac{\tilde{l}_{\text{eff}}}{l_{\text{mc}}}, \quad (\text{S11})$$

where we use Matthiessen’s rule $l^{-1} = l_{\text{mc}}^{-1} + l_{\text{mr}}^{-1}$ to combine the depopulation of electronic states with a combined mean free path l . Momentum-conservation is accounted for by the last term, which defines the ‘average’ mean free path $\tilde{l}_{\text{eff}}(y)$

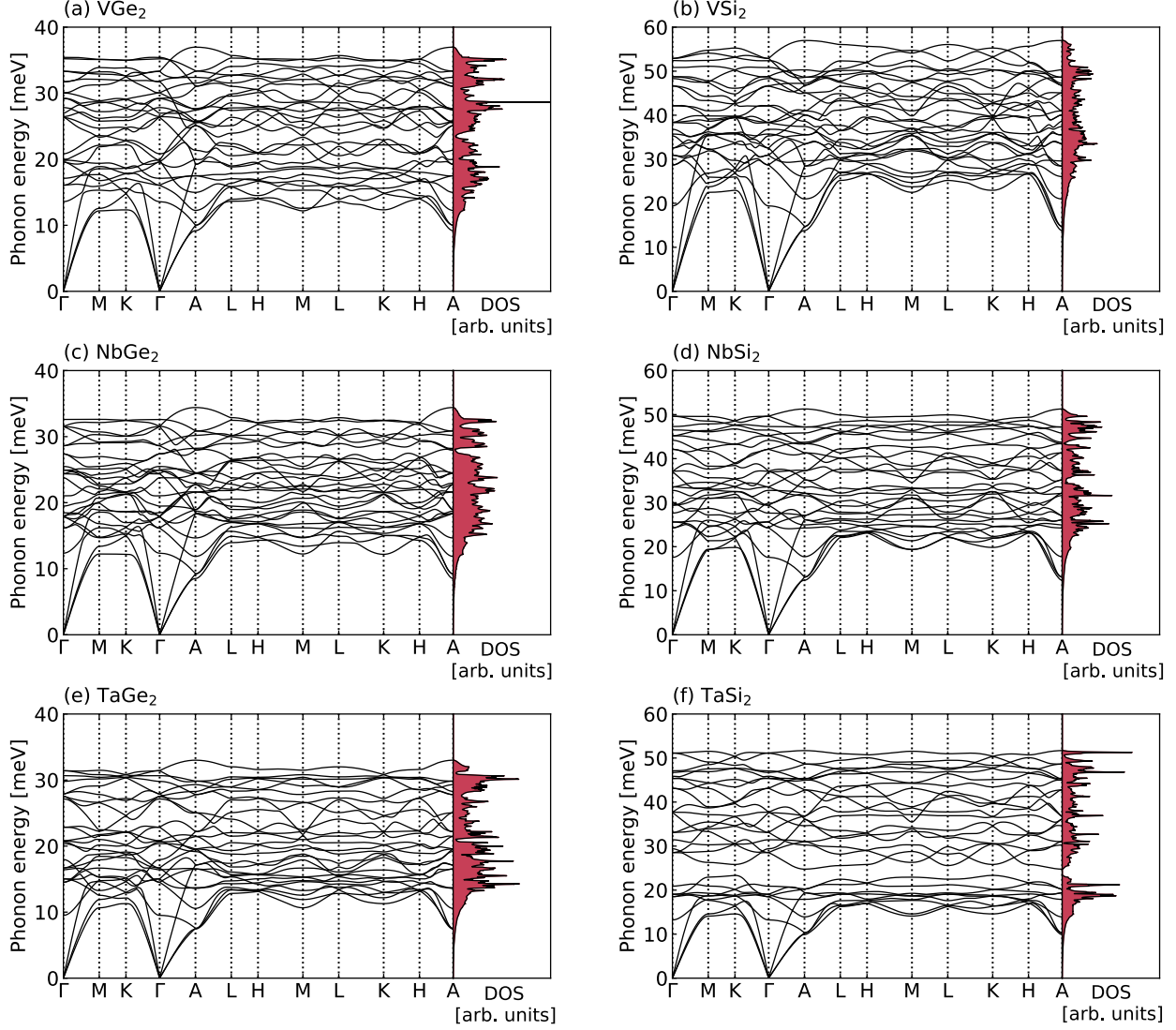


Fig. S5. *Ab initio* phonon spectra and phonon density of states for all six transition metal ditellurides that crystallize in space group 180 or 181.

and is directly proportional to current density, $j_x(y)$:

$$\tilde{l}_{\text{eff}}(y) = \int_0^{2\pi} \frac{d\theta}{\pi} \cos^2(\theta) l_{\text{eff}}(y, \theta) \quad (\text{S12})$$

$$j_x(y) = \left(\frac{m}{\pi \hbar^2} \right) \varepsilon_F e^2 \frac{E_x}{m v_F} \tilde{l}_{\text{eff}}(y). \quad (\text{S13})$$

where m is the electronic effective mass, E_x is the electric field component in the x direction, and v_F is the Fermi velocity. Equation (S11) is solved numerically by transforming it into a Fredholm integral equation of the 2nd kind following Refs. [5, 6]. For each pair of (l_{mr}/W , l_{mc}/W) values, solution of eqs. (S11) to (S13) returns the spatially-resolved current density profile $j_x(y)$. From this, we extract the scalar-valued current density curvature, by fitting a parabola around the center of the channel $y=0$. Phase diagrams produced using this metric (Fig. 4b) are by construction normalized, with fully-parabolic (‘Poiseuille’) flow having a curvature equal to unity and full-diffusive (‘Ohmic’) flow having a curvature equal to zero.

COMPUTATIONAL DETAILS

For the electron-phonon coupling, lifetime and resistivity calculations, we first obtain the electronic structure of each group V transition metal ditetrelide from first principles using density functional theory (DFT) as implemented in JDFTx [7]. We use fully relativistic optimized norm-conserving Vanderbilt pseudopotentials [8–10] for NbSi₂, NbGe₂, VSi₂, and VGe₂ and fully relativistic Rappe-Rabe-Kaxiras-Joannopoulos ultrasoft (RRKJUS) pseudopotentials [11, 12] for TaSi₂ and TaGe₂, all for the PBEsol exchange-correlation functional [13]. For all compounds, we set a uniform $14 \times 14 \times 10$ k -point mesh across each 9-atom standard primitive unit cell, plane-wave energy cutoffs of 41 Hartrees, and Marzari-Vanderbilt (“cold”) smearing [14] with a 0.001 Hartree width. The lattice constants and ionic positions of each material are relaxed such that the forces on all atoms are less than 10^{-9} Ha/Bohr. We employ a frozen phonon approximation with a $2 \times 2 \times 2$ supercell to calculate each set of phonon states, and construct bases of 78 maximally localized Wannier functions (MLWFs) [15, 16] from the DFT plane-wave electronic states, which allow us to accurately interpolate electron and phonon properties to denser corresponding $\mathbf{k}$ and $\mathbf{q}$ meshes for effective Brillouin zone integration [17]. For the Coulomb-mediated electron-electron scattering lifetimes of NbGe₂ and NbSi₂, dielectric matrix cutoffs of 200 eV were used with an energy resolution of 0.1 eV.

KOHN-SHAM BANDS AND PHONON SPECTRA IN TRANSITION METAL DITETRELIDES

To complement the main text, we show the single-electron band structure (Fig. S4) obtained as outlined above, as well as the phonon spectra and their density of states (Fig. S5) for all six transition metal ditetrelides that crystallize in space group 180 or 181.

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